Gravitational potential, particles and halos: GADGET-4 versus PKDGRAV3 from identical initial conditions

32^3 and 256^3 particles in $200\,h^{-1}$ Mpc, Sahyadri default settings on pawna

Sahyadri sandbox, branch `gadget-pkdgrav3-potential`

11 November 2026

Abstract

Gadget-4 and pkdgrav3 are run from identical initial conditions (Gadget-4 NGenIC, Sahyadri default settings, $200\,h^{-1}$ Mpc) at 32^3 and 256^3 particles, and compared particle by particle (potential, position, velocity), halo by halo (Rockstar, consistent-trees, VAHC) and through the stacked potential around halos in mass bins out to $50\,h^{-1}$ Mpc. At 32^3 the potentials agree to 2–3% rms until $z \approx 3$ and to 25% at $z = 0$, most of which comes from the particles ending up in different places (median separation $0.38\,h^{-1}$ Mpc at $z = 0$); the zero points agree to $2 \times 10^3\,(\mathrm{km\,s^{-1}})^2$ once pkdgrav3's self-term is removed; the stacked potential around the most massive halos agrees within errors beyond $\sim 1\,h^{-1}$ Mpc and is deeper in pkdgrav3 at small radii (softening). **Status (11 Nov 2026, 19:50): the 256^3 runs are in progress on pawna; Section 5 will be filled in when they finish.**

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1 Set-up

Both codes evolve the *same* particle load. GADGET-4's NGenIC (2LPT, CLASS $P(k)$ at $z = 49$, the Sahyadri production recipe of `scripts/submit_CiPod.sh`) writes the initial conditions; the same file, converted to GADGET-2 format, is read by PKDGRAV3 (`achInFile`), so the two runs differ only in the N-body solver, the time stepping and the softening. Every setting is the Sahyadri default of `config/submit/sahyadri/sub_default.conf` unless stated (Table 1).

box, particles	$L = 200\,h^{-1}$ Mpc; $N = 32^3$ (test) and 256^3
cosmology	$\Omega_m = 0.3137721$, $\Omega_b = 0.0493017$ (GADGET-4 parameter only; both runs are dark matter only), $\Omega_\Lambda = 0.6862279$, $h = 0.6736$, $n_s = 0.9649$, $A_s = 2.098903 \times 10^{-9}$, seed 12345678
initial conditions	GADGET-4 NGenIC 2LPT at $z = 49$ (CLASS spectrum, no renormalisation)
outputs	Sahyadri's 100 output times (`config/sims/outputs100.txt`, $z = 12 \to 0.0093$) plus $a = 1$: 101
GADGET-4	commit 1e171a4, production Config + `EVALPOTENTIAL`; TreePM, PMGRID $= 2N^{1/3}$, `ErrTolForceAcc` 0.002, $\theta_{\max} = 0.75$; softening $\epsilon_G = L/(30N^{1/3})$ (Plummer-equivalent)
PKDGRAV3	`shadaba/pkdgrav3` `main` (binary built from 5766bb28; later commits are documentation); production `sahyadri2048.par` settings: $\epsilon_P = 1/(60N^{1/3})$ box units, 20 steps between outputs, output at the redshifts GADGET-4 actually wrote; `dCompPotStep=1` (potential in the `.pkc` files)
softening	256^3: $\epsilon_G = 26.0\,h^{-1}$kpc, $\epsilon_P = 13.0\,h^{-1}$kpc ($32^3$: 208 and $104\,h^{-1}$kpc); PKDGRAV3's force is Newtonian beyond $2\epsilon_P = \epsilon_G$, GADGET-4's (spline) beyond $2.8\,\epsilon_G$
halos	Rockstar (production template: 30 particles output, 20 minimum, unbound threshold 0.5; GADGET-4 HDF5 via `FILE_FORMAT = "AREPO"`, PKDGRAV3 via the native `.pkc` reader), consistent-trees, VAHC (grid 512^3)

Table 1: Simulation and pipeline settings.

Pipeline. One command submits everything as a PBS chain (`scripts/gadget_pkdgrav3_compare/submit_compare.sh compare256.conf`): GADGET-4 job (CLASS → NGenIC → IC conversion → run) → PKDGRAV3 `sim` → Rockstar → consistent-trees → VAHC for each code (the Sahyadri PKDGRAV3 pipeline, `scripts/pkdgrav3_pipeline`) → comparison job (`job_compare.pbs`). At 32^3 consistent-trees cannot run (at most 15 halos per snapshot; it needs enough halos to estimate its consistency tolerances), so the 32^3 halo results use the Rockstar catalogues.

2 Where things are

Data on pawna

Work directory `/storage/shadab/pkdgrav3_io/potcmp/` (below: `data/` = `/storage/shadab/pkdgrav3_io/potcmp/data/`).

what	path
GADGET-4 256^3 run	`data/sims/potcmp/gadget256/r1/`: `snapshot_000..100.hdf5` (with Potential), `param_run.txt`, `gadget_run.log`, `snapnames.txt`, `gadget_out_z.txt`
initial conditions 256^3	`data/sims/potcmp/gadget256/r1/ics/`: `snapshot_ics_000.hdf5` (GADGET-4), `ics_gadget2.dat` (PKDGRAV3 input), `param_ics.txt`; CLASS: `data/transfer/class_gadget256*`
PKDGRAV3 256^3 run	`data/sims/potcmp/pkd256/r1/`: `p.par`, `snap.<step>` (HDF5), `snap.<step>.s<pct>.0.pkc`, `snapnames.txt`, `run.log`
halos 256^3	`data/halos/potcmp/gadget256/r1/`, `data/halos/potcmp/pkd256/r1/`: `out_<snap>.list` (Rockstar), `out_<snap>.trees` (consistent-trees hlists), `out_<snap>.vahc`, `trees/`, `hlists/`; `../scales.txt`
comparison 256^3	`data/compare/pkd256_vs_gadget256_r1/`: `particles.csv`, `halos.csv`, `summary.json`, `snap_*.npz`, `profiles/{gadget,pkd,pkc}_out<i>.npz` (stacks), `halos/` (`halo_mf.csv`, `halo_match.csv`, `halo_match_out<i>.npz`), logs
32^3 (same layout)	`data/sims/potcmp/{gadget32,pkd32}/r1/`, `data/halos/potcmp/{gadget32,pkd32}/r1/`, `data/compare/pkd32_vs_gadget32_r1/` (stacks in `profiles/`; the earlier $r/R_{\rm vir}$ profiles in `profiles_rvir_old/`)
first 32^3 run (no `EVALPOTENTIAL`)	`/storage/shadab/pkdgrav3_io/potcmp/data_run1_zeropot/` (Potential $= 0$; kept as evidence)
job logs	`data/logs/<stage>.potcmp_{gadget,pkd}256_r1.<job>.log`, chains `data/logs/chain.*.txt`
GADGET-4 builds	`/storage/shadab/pkdgrav3_io/potcmp/g4home/code/Gadget-4/{mesh512, mesh512-NGenIC,mesh64,mesh64-NGenIC}/Gadget4`
PKDGRAV3 binary	`/storage/shadab/pkdgrav3_io/build/pkdgrav3`

Scripts and report (repository `a-paranjape/sahyadri-sandbox`, branch `gadget-pkdgrav3-potential`)

`scripts/gadget_pkdgrav3_compare/`	
`compare32.conf`, `compare256.conf`, `*_gadget.conf`	run configurations (one per resolution; the `_gadget` file switches to the GADGET-4 side)
`submit_compare.sh`	submits the whole chain
`job_gadget.pbs`, `setup_gadget4_pawna.sh`, `Config.sh.template`, `make_gadget4_param.py`	GADGET-4 build, ICs and run, Sahyadri recipe
`gadget4_ic_to_gadget2.py`	the GADGET-4 ICs for PKDGRAV3 (IDs $\to$ iOrder, mass so that PKDGRAV3's total is Ω_m)
`job_compare.pbs`	comparison job: the three scripts below
`compare_potential.py`	particle by particle: potential, positions, velocities; FFT Poisson reference
`halo_potential_profile.py`	stacked potential around halos in mass bins (Section 3)
`compare_halos.py`	mass functions, matched halos, VAHC quantities
`reports/gadget_pkdgrav3_potential/`	this report: `report.tex`, `stack_potential.ipynb` (stacked potential figures), `make_figures.py` (other figures), `figs/`, `data/` (copies of the small comparison outputs: `particles.csv`, `profiles/`, `halos/*.csv`)

3 Methods

Units and zero point of the potential. All potentials are peculiar and physical, in $(\mathrm{km\,s^{-1}})^2$. GADGET-4: $\phi =$ `Potential`$/a$ (attribute `a_scaling` $= -1$; the self-interaction is excluded and the periodic potential has volume mean zero). PKDGRAV3: $\phi =$ `Potential`$\,U^2/a$ with $U =$

$100\,L/\sqrt{8\pi/3}\,\mathrm{km\,s^{-1}}$ ($G = 1$, $\rho_{\rm crit} = 3/8\pi$, $L = 1$); PKDGRAV3 includes each particle's softened self-interaction $-\frac{105}{64}\,m/\epsilon_P$, a constant for equal masses, which is removed. The two codes can still differ by a constant per snapshot (zero point of the periodic sum, e.g. the Madelung-like term of a particle's own images). It changes no force, so it is measured as the median of $\phi_{\rm pkd} - \phi_{\rm Gadget}$ over all particles and removed ("aligned"). The `.pkc` potential is PKDGRAV3's, quantised to a step of $1\,(\mathrm{km\,s^{-1}})^2$ (maximum error 0.5, checked at every output).

Velocities. Peculiar, $\mathrm{km\,s^{-1}}$: GADGET-4 `Velocities` $\sqrt{a}$, PKDGRAV3 `Velocities` Ua (at $z = 12$ the two rms velocities agree to 0.4%).

Particles are matched by ID (GADGET-4 ID -1 = PKDGRAV3 iOrder). For every 5th output (every output at 32^3) `compare_potential.py` reports the correlation and rms difference of ϕ, $|\Delta\vec{x}|$ and $|\Delta\vec{v}|$ percentiles, and, as an independent reference, the potential from an FFT Poisson solution of each snapshot's own particles (CIC on $\min(4N^{1/3}, 512)^3$, deconvolved). The rms difference between the FFT potentials of the two codes' particle positions isolates the part of $\Delta\phi$ that comes from the particles being in different places rather than from the force solver.

Stacked potential profiles. `halo_potential_profile.py` computes, for host halos (`pid` $= -1$) in bins of $M_{\rm vir}$, the mean $\langle\phi\rangle(r)$ and standard deviation $\sigma_\phi(r)$ of the potential of all (halo, particle) pairs in logarithmic shells of comoving distance r from 0.02 to $50\,h^{-1}\,$Mpc (36 bins), and the jackknife error of the mean over 3^3 sub-cubes of halo centres. Each code uses its own halos and its own particles. Only stacks are accumulated: per chunk of particles, a periodic KD-tree of the chunk is paired with trees of halo centres, and every pair adds $(1, \phi, \phi^2, r)$ to a (sub-cube, mass bin, shell) array, so memory does not depend on the number of halos or particles. To keep the cost per halo constant out to large radii the outer shells are diluted: shell i uses the nested random subsample $u < 8^{-\ell_i}$ of the particles, with the smallest ℓ_i for which the expected number of particles inside the shell's outer radius is ≤ 4096. Each shell uses one level, so its mean and scatter are unbiased (checked against a brute-force calculation: identical without dilution, within $1\sigma_{\rm jk}$ with it). The pair counts also give the halo–matter cross-correlation $1 + \xi_{hm}(r)$. At 2048^3 the outermost level uses 8^{-6} of the particles, and the cost is $\sim n_{\rm halo}\times$ levels $\times 4096$ pairs plus one pass through the snapshot. The figures are made by the notebook `stack_potential.ipynb`.

Halo catalogues. `compare_halos.py` compares host-halo mass ($M_{\rm vir}$, M_{200b}) and $V_{\rm max}$ functions, and matches host halos one to one (mutual nearest centres within $0.5\,R_{\rm vir}$ of the GADGET-4 halo and $|\log_{10} M_{\rm pkd}/M_{\rm Gadget}| < 0.5$). For matched halos it reports, per mass bin, the median and 16–84% range of the ratios of $M_{\rm vir}$, M_{200b}, $V_{\rm max}$, concentration $R_{\rm vir}/r_s$, spin, shape and virial ratio, the centre offset and bulk-velocity difference, and the differences of the VAHC halo bias b_1 and tidal eigenvalues.

4 Results at 32^3

The 32^3 run is the first test of the chain (particle mass $2.1 \times 10^{13}\,h^{-1}\,\mathrm{M_\odot}$, at most 15 halos per snapshot). Table 2 and the grey lines of Fig. 3 summarise the particle comparison.

out	z	corr(ϕ)	$\frac{\text{rms}\,\Delta\phi}{\text{rms}\,\phi}$	FFT$_x$	offset $(\text{km s}^{-1})^2$	med $\lvert\Delta x\rvert$ h^{-1} Mpc	90% $\lvert\Delta x\rvert$ h^{-1} Mpc	med $\lvert\Delta v\rvert$ km s^{-1}	corr(v)
0	11.98	0.99965	0.026	0.082	+1392	0.016	0.029	6.3	0.9989
19	2.96	0.99972	0.024	0.103	+451	0.098	0.167	13.3	0.9980
46	1.00	0.99520	0.098	0.117	+308	0.205	0.364	17.9	0.9633
64	0.50	0.98510	0.173	0.147	−419	0.271	0.531	20.8	0.9123
100	0.00	0.96891	0.253	0.220	−2011	0.379	0.891	24.9	0.8151

Table 2: 32^3, particle by particle. FFT$_x$: rms difference of the FFT potentials computed from the two codes' particle positions, over rms ϕ. Offset: median $\phi_{\text{pkd}} - \phi_{\text{Gadget}}$ after removing PKDGRAV3's self-term (-1.9×10^7 at $z = 12$, -1.4×10^6 $(\text{km s}^{-1})^2$ at $z = 0$).

- **Potential.** Until $z \approx 3$ the two potentials agree to 2–3% rms (correlation 0.9997). Afterwards the difference grows to 25% at $z = 0$; the FFT potentials of the two codes' own particle positions differ by 22%, so most of the late-time difference comes from the particles being in different places, not from the force solvers. The zero points agree to $\lesssim 2 \times 10^3$ $(\text{km s}^{-1})^2$ (0.1% of the self-term that PKDGRAV3 carries).

- **Positions and velocities.** The median particle separation grows from $16\,h^{-1}$kpc at $z = 12$ to $0.38\,h^{-1}$ Mpc at $z = 0$ ($1.8\,\epsilon_G$, 6% of the mean interparticle spacing $6.25\,h^{-1}$ Mpc), the median velocity difference from 6 to 25 km s^{-1}. The rms velocity difference (314 km s^{-1} at $z = 0$) is dominated by particles in halos, where trajectories are chaotic: there the median is small but the tail long (Fig. 1). The rms velocities of the two runs differ by 0.4% at $z = 12$ and 1.7% at $z = 0$.

- **Halos.** Rockstar finds 13 host halos in GADGET-4 and 8 in PKDGRAV3 at $z = 0$; 4 are matched one to one (the others merge or split differently at the 30–40 particle limit). Matched masses agree to a few per cent.

- **Stacked potential** (Fig. 2). Around the 9 halos above $4 \times 10^{14}\,h^{-1}\,\text{M}_\odot$ the mean potential of the two codes agrees within the jackknife errors from 1 to $50\,h^{-1}$ Mpc; inside $\sim 0.5\,h^{-1}$ Mpc PKDGRAV3's is deeper, as expected from its smaller effective softening (PKDGRAV3's force becomes Newtonian at $1.0\,\epsilon_G$, GADGET-4's at $2.8\,\epsilon_G$). With 1–9 halos per bin these numbers are indicative only.

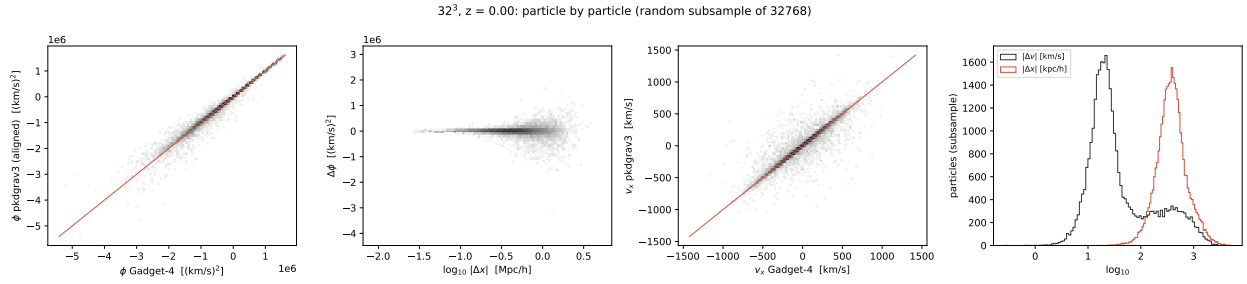


Figure 1: 32^3, $z = 0$, every particle: potentials (PKDGRAV3 aligned to GADGET-4's zero point), potential difference against particle separation, v_x, and the distributions of $|\Delta v|$ and $|\Delta x|$.

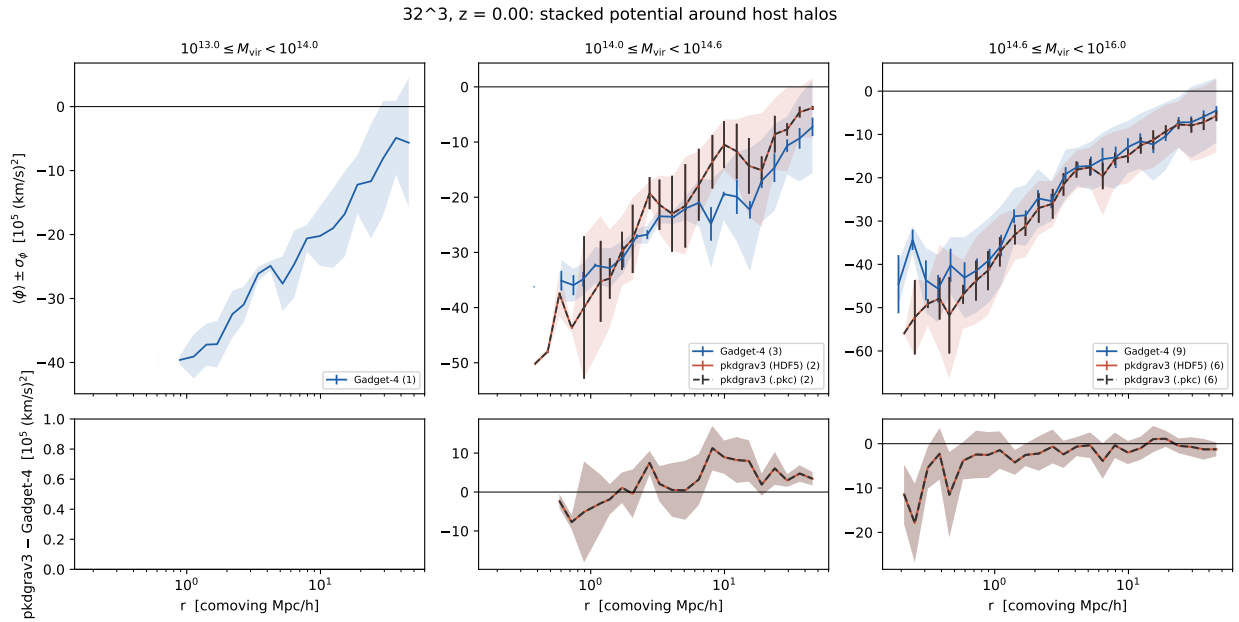


Figure 2: 32^3, $z = 0$: stacked potential around host halos in three $M_{\rm vir}$ bins (number of halos in the legend). Line with error bars: mean and jackknife error; band: $\pm\sigma_\phi$, the scatter of the potential over all pairs in the shell. Bottom: PKDGRAV3 − GADGET-4 (HDF5 and `.pkc` coincide).

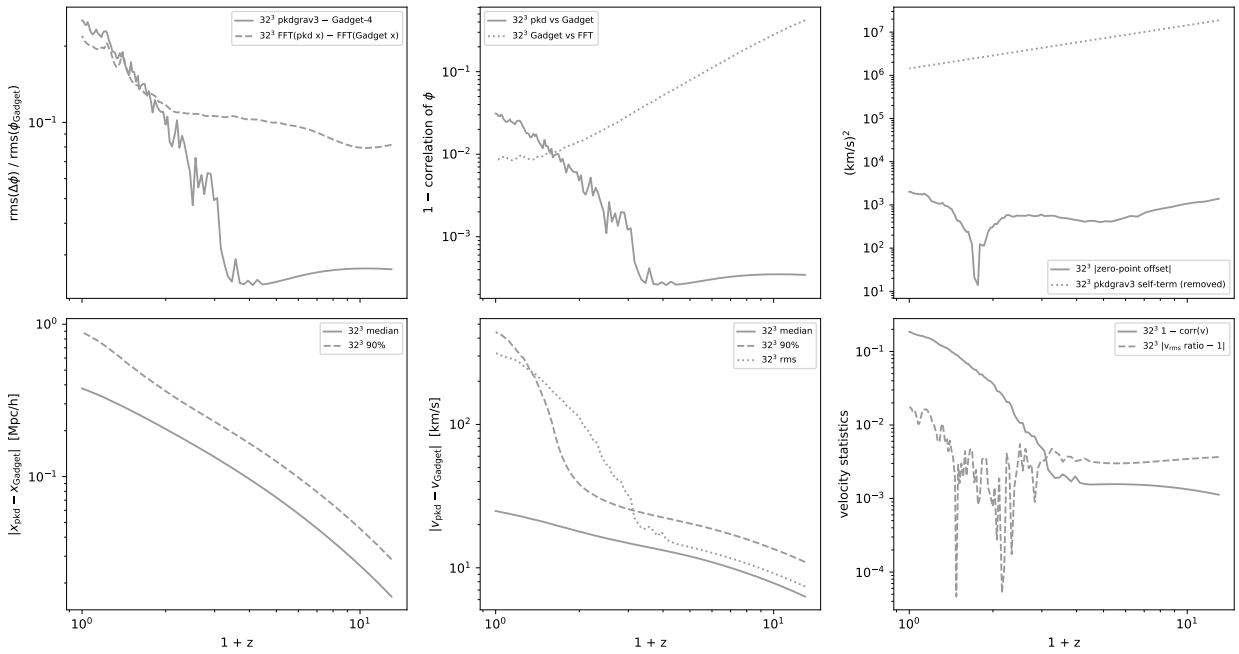


Figure 3: Particle-by-particle statistics against redshift (grey: 32^3; the 256^3 run is added when it finishes). Top: rms potential difference over rms potential (solid) and the same for the FFT potentials of the two codes' particle positions (dashed); 1−correlation of the potentials; zero-point offset and PKDGRAV3's removed self-term. Bottom: particle separation, velocity difference, velocity correlation and rms-velocity ratio.

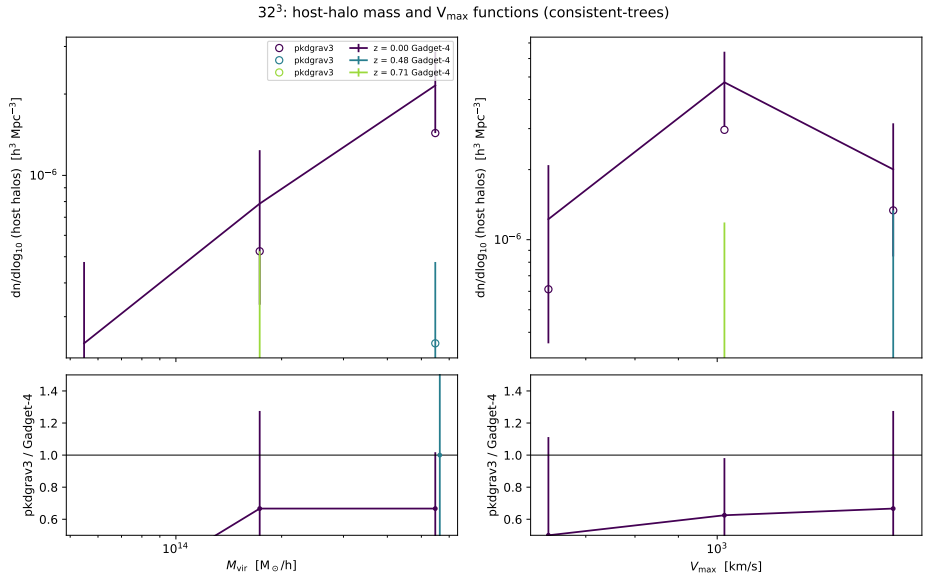


Figure 4: 32^3: host-halo $M_{\rm vir}$ and $V_{\rm max}$ functions (Rockstar) of both codes; with ≤ 15 halos per snapshot this is only a test of the tool.

5 Results at 256^3

In progress. The 256^3 chain (PBS jobs 2536537–2536545) was submitted on 11 November 2026 at 19:15: GADGET-4 run → PKDGRAV3 run → Rockstar, consistent-trees and VAHC for both codes → comparison. Outputs will appear in the paths of Section 2; this section will contain the particle comparison (potential, positions, velocities), the stacked potential in mass bins (2×10^{12}–$10^{16}\,h^{-1}\,\mathrm{M}_\odot$, 0.02–$50\,h^{-1}\,$Mpc), mass and $V_{\rm max}$ functions, matched-halo properties and the VAHC quantities.

6 Problems found on the way

1. **Gadget-4 writes Potential = 0 without `EVALPOTENTIAL`.** The first 32^3 run, with the exact production Config (`OUTPUT_POTENTIAL` only), has zero potential everywhere (`data_run1_zero pot/`). The production Sahyadri Config lacks `EVALPOTENTIAL`, so the production GADGET-4 snapshots most likely have no potential (not checked directly). Adding the flag left the ICs and output times byte-identical.

2. **consistent-trees needs many halos per snapshot**: it fails at 32^3 ("too few halos at scale factor 0.768278 to calculate consistency metric") even after relaxing `gen_merger_cfg.pl`'s 20000-byte catalogue threshold. At 256^3 the production threshold is used.

3. **pkdgrav3 has no physical units when it reads an IC file** (`units.cxx`): box size in the output headers and `.pkc` potential units are wrong unless `dKpcUnit` and `dMsolUnit` are given; the pipeline's parameter generator now writes them. A Gadget-2 input file's IDs become iOrder directly, so they must be $0 \ldots N-1$; PKDGRAV3's HDF5 `Header/Time` is code time, not a. Recorded in the PKDGRAV3 fork, `docs/compressed_io_plan.md` section 7.

7 Reproducing

```
# on pawna, in a clone of sahyadri-sandbox (branch gadget-pkdgrav3-potential)
cd scripts/gadget_pkdgrav3_compare
bash -lc "bash setup_gadget4_pawna.sh <gadget4 source> <G4HOME> 256"   # Gadget-4 builds (onc
```

```
bash submit_compare.sh compare256.conf                                  # 9 PBS jobs
# figures (Mac or anywhere with matplotlib), from reports/gadget_pkdgrav3_potential:
rsync pawna:<data>/compare/pkd256_vs_gadget256_r1/{particles.csv,profiles,halos} data/pkd256_
jupyter nbconvert --to notebook --execute stack_potential.ipynb
python make_figures.py 32=data/pkd32_vs_gadget32_r1 256=<copy with snap_*.npz>
latexmk -pdf report.tex
```